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REPORT ON
EQUITY VALUATION
INDIAN FMCG COMPANIES

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Contents

1 Valuation Methodology: Net Asset Value (Intrinsic Value) Method

1.1 Theoretical Framework and Valuation Premise

In corporate financial accounting and quantitative equity valuation, the **Net Asset Value (NAV) Method**—also formally referred to as the **Liquidation Value Method** or **Asset-Backing Method**—determines the intrinsic equity value of an enterprise strictly through the prism of its tangible balance sheet resources.

Under the classical accounting doctrine, the intrinsic value of a firm represents the residual financial claim that equity shareholders would legally receive if all company assets were liquidated at fair market value and all prior creditor obligations, debt securities, and external liabilities were settled in full.

Definition 1.1: Net Asset Value (Intrinsic Value)

The **Net Asset Value** available to equity shareholders is defined as the aggregate fair value of all tangible assets less total external obligations. The resulting figure, when normalized across the total count of outstanding common shares, constitutes the **Intrinsic Value per Share** (IV):

$$\text{IV} = \frac{\text{Net Assets Available to Equity Shareholders}}{\text{Number of Equity Shares Outstanding}} \tag{1}$$

1.2 Mathematical Formulation

1.2.1 Goodwill Estimation via the Purchase of Average Profits Method

Goodwill represents the capitalized value of super-normal earning power. Under the canonical accounting valuation method, goodwill is valued at **three years' purchase of the average annual net profits** realized over the preceding five accounting periods:

$$\overline{\Pi} = \frac{1}{5}\sum_{t=1}^{5} \Pi_t \tag{2}$$

where Π_t denotes the profit after tax for fiscal year t. The capitalized goodwill is subsequently evaluated as:

$$\text{Goodwill} = \overline{\Pi} \times N_{\text{purchase}} = \overline{\Pi} \times 3 \tag{3}$$

1.2.2 Net Assets Available to Equity Shareholders

In empirical equity research, observable secondary market liquidation values for plant, property, equipment, and inventories are rarely disclosed in quarterly balance sheets. Consequently, audited balance sheet book values are adopted as the standard regulatory proxy.

To maintain strict accounting conservatism and isolate the liquidation floor, existing unamortized or recorded goodwill is **excluded** from total assets, ensuring that only tangible resources back the equity:

$$\text{Tangible Assets} = \text{Total Assets} - \text{Goodwill} \tag{4}$$

$$\text{Net Assets} = \text{Tangible Assets} - \text{Total Liabilities} \tag{5}$$

Equivalently written in a single unified formulation:

$$\text{Net Assets} = \text{Total Assets at Market/Book Value} - \text{Goodwill} - \text{Total Liabilities} \tag{6}$$

1.2.3 Intrinsic Value per Share

Dividing the aggregate net tangible assets by the total volume of paid-up equity shares yields the per-share intrinsic worth:

$$\boxed{\mathrm{IV} = \frac{\text{Net Assets}}{\text{Equity Shares Outstanding}}} \tag{7}$$

1.2.4 Market Divergence and Overvaluation Metrics

To measure how the prevailing open-market quotation diverges from its underlying asset-backed intrinsic valuation, two formal metrics are defined:

(a) Market Price per Share (P_0):

$$P_0 = \frac{\text{Market Capitalization}}{\text{Number of Equity Shares Outstanding}} \tag{8}$$

(b) Absolute Valuation Gap (Δ):

$$\Delta = \mathrm{IV} - P_0 \tag{9}$$

A negative valuation gap ($\Delta < 0$) indicates that the current market price trades at a substantial premium above the tangible asset base, signaling overvaluation under the liquidation framework.

(c) Overvaluation Multiple / Ratio ($\mathcal{R}$):

$$\mathcal{R} = \frac{P_0}{\mathrm{IV}} = \frac{\text{Market Capitalization}}{\text{Net Assets}} \tag{10}$$

The ratio $\mathcal{R}$ quantifies the multiple of tangible net assets that investors in public capital markets are willing to pay for every unit of physical equity backing.

1.3 TikZ Process Flowchart: Net Asset Value Pipeline

Figure 1 details the operational filtration sequence of the Net Asset Value methodology from consolidated corporate assets down to intrinsic value and the comparative valuation gap.

2 Empirical Valuation of Ten Leading Indian FMCG Companies

2.1 Cross-Sectional Dataset and Balance Sheet Composition

To assess the empirical behavior of the Net Asset Value method across the Indian Fast-Moving Consumer Goods (FMCG) sector, a representative sample of ten prominent blue-chip corporations was analyzed. Financial metrics are calibrated in Crores of Indian Rupees (Rs. Cr), with per-share values expressed in Rupees (Rs.).

Table 1 provides the granular breakdown of total assets, recorded goodwill deductions, tangible assets, total liabilities, net assets available to shareholders, and the resulting intrinsic value per share.

2.2 Worked Valuation Walkthroughs

To demonstrate the exact algebraic steps prescribed by the valuation framework, detailed step-by-step calculations are presented for two polar cases: **ITC Ltd** (a high tangible-asset enterprise) and **Nestle India Ltd** (a lean, asset-light multinational).

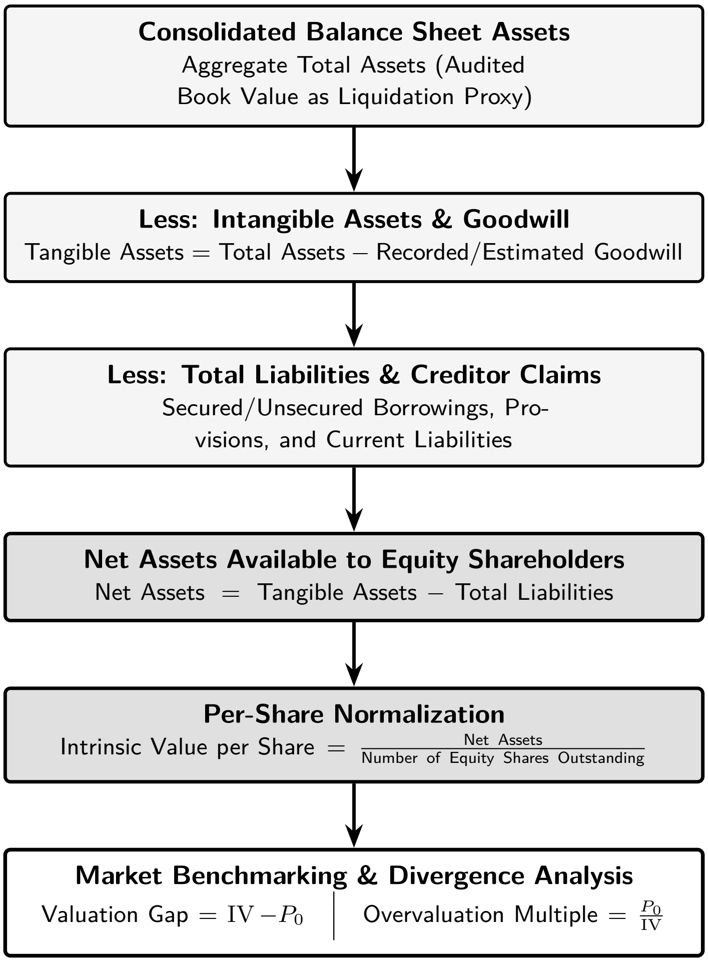

Figure 1: Net Asset Value (Intrinsic Value) Quantitative Filtration Pipeline.

Company Name	**Total Assets** (Rs. Cr)	**Goodwill** (Rs. Cr)	**Tangible Assets** (Rs. Cr)	**Total Liab.** (Rs. Cr)	**Net Assets** (Rs. Cr)	**Equity Shares** (Cr)	**Intrinsic Value** (Rs.)
Hindustan Unilever Ltd	79,738	1,478	78,260	30,999	47,261	235	**201**
ITC Ltd	93,637	2,399	91,238	16,981	74,257	1,253	**59**
Nestle India Ltd	13,357	444	12,913	6,641	6,272	193	**33**
Varun Beverages Ltd	25,541	0	25,541	7,639	17,902	676	**26**
Britannia Industries Ltd	9,730	1,380	8,350	4,648	3,702	24	**154**
Marico Ltd	9,950	557	9,393	5,183	4,210	130	**32**
Tata Consumer Products	34,279	2,820	31,459	11,264	20,195	99	**204**
Godrej Consumer Products	13,365	2,948	10,417	5,573	4,844	102	**47**
Dabur India Ltd	17,480	1,287	16,193	6,238	9,955	177	**56**
Colgate-Palmolive India	3,408	47	3,361	1,394	1,967	27	**73**

Table 1: Comprehensive Net Asset Value and Intrinsic Value Computation for FMCG Firms.

Example 2.1: Valuation Walkthrough: ITC Limited

Balance Sheet Inputs (in Rs. Crores):

$$\begin{aligned} \text{Total Assets} &= \text{Rs. } 93{,}637\,\text{Cr} \\ \text{Goodwill (Deducted)} &= \text{Rs. } 2{,}399\,\text{Cr} \\ \text{Total Liabilities} &= \text{Rs. } 16{,}981\,\text{Cr} \\ \text{Outstanding Shares} &= 1{,}253 \text{ Cr shares} \end{aligned}$$

Step 1: Determine Net Tangible Assets Available to Shareholders:

$$\begin{aligned} \text{Tangible Assets} &= \text{Total Assets} - \text{Goodwill} = 93{,}637 - 2{,}399 = \text{Rs. } 91{,}238\,\text{Cr} \\ \text{Net Assets} &= \text{Tangible Assets} - \text{Total Liabilities} = 91{,}238 - 16{,}981 = \text{Rs. } 74{,}257\,\text{Cr} \end{aligned}$$

Step 2: Normalization to Per-Share Intrinsic Worth:

$$\text{IV}_{\text{ITC}} = \frac{\text{Net Assets}}{\text{Shares}} = \frac{74{,}257}{1{,}253} \approx \text{Rs. } 59.26 \implies \text{Rs. } \mathbf{59}$$

Example 2.2: Valuation Walkthrough: Nestle India Limited

Balance Sheet Inputs (in Rs. Crores):

$$\begin{aligned} \text{Total Assets} &= \text{Rs. } 13{,}357\,\text{Cr} \\ \text{Goodwill (Deducted)} &= \text{Rs. } 444\,\text{Cr} \\ \text{Total Liabilities} &= \text{Rs. } 6{,}641\,\text{Cr} \\ \text{Outstanding Shares} &= 193 \text{ Cr shares} \end{aligned}$$

Step 1: Compute Tangible Net Asset Pool:

$$\begin{aligned} \text{Tangible Assets} &= 13{,}357 - 444 = \text{Rs. } 12{,}913\,\text{Cr} \\ \text{Net Assets} &= 12{,}913 - 6{,}641 = \text{Rs. } 6{,}272\,\text{Cr} \end{aligned}$$

Step 2: Derive Intrinsic Value per Share:

$$\text{IV}_{\text{Nestle}} = \frac{6{,}272}{193} \approx \text{Rs. } 32.50 \implies \text{Rs. } \mathbf{33}$$

2.3 Market Price Comparison and Valuation Gap Analysis

Table 2 tabulates the comparison between the computed intrinsic value and the prevailing market price per share, alongside the resulting valuation gap $\Delta = \mathrm{IV} - P_0$.

Company Name	**Intrinsic Value (Rs.)**	**Market Price (Rs.)**	**Valuation Gap (Rs.)**	**Valuation Status**
Hindustan Unilever Ltd	201	1,973	$-1,772$	Highly Overvalued
ITC Ltd	59	264	-205	Highly Overvalued
Nestle India Ltd	33	1,409	$-1,376$	Highly Overvalued
Varun Beverages Ltd	26	204	-178	Highly Overvalued
Britannia Industries Ltd	154	5,120	$-4,966$	Highly Overvalued
Marico Ltd	32	814	-782	Highly Overvalued
Tata Consumer Products	204	1,010	-806	Highly Overvalued
Godrej Consumer Products	47	881	-834	Highly Overvalued
Dabur India Ltd	56	382	-326	Highly Overvalued
Colgate-Palmolive India	73	1,843	$-1,770$	Highly Overvalued

Table 2: Market Price vs. Intrinsic Value and Resulting Valuation Gaps.

2.4 Overvaluation Ratio Hierarchy

Table 3 sorts all ten companies in ascending order of their Overvaluation Ratio $\mathcal{R} = P_0/\,\mathrm{IV}$. This reveals the spectrum of pricing premiums prevailing in the public equity market relative to tangible liquidation values.

Company Name	**Market Price (Rs.)**	**Intrinsic Value (Rs.)**	**Overvaluation Ratio**
ITC Ltd	264	59	**4.5**×
Tata Consumer Products	1,010	204	**4.9**×
Dabur India Ltd	382	56	**6.8**×
Varun Beverages Ltd	204	26	**7.8**×
Hindustan Unilever Ltd	1,973	201	**9.8**×
Godrej Consumer Products	881	47	**18.7**×
Colgate-Palmolive India	1,843	73	**25.3**×
Marico Ltd	814	32	**25.4**×
Britannia Industries Ltd	5,120	154	**33.2**×
Nestle India Ltd	1,409	33	**42.7**×

Table 3: Ranked Overvaluation Multiples ($P_0/\,\mathrm{IV}$) Across Sample Companies.

3 Cross-Sectional Analysis and Margin of Safety Spectrum

3.1 Visualizing the Overvaluation Spectrum

A cross-sectional examination of the ten sampled companies reveals stark dispersion in their respective market premiums. Figure 2 charts the overvaluation multiple ($P_0/\,\mathrm{IV}$) across all firms, providing an immediate visual benchmarking of asset-backing strength.

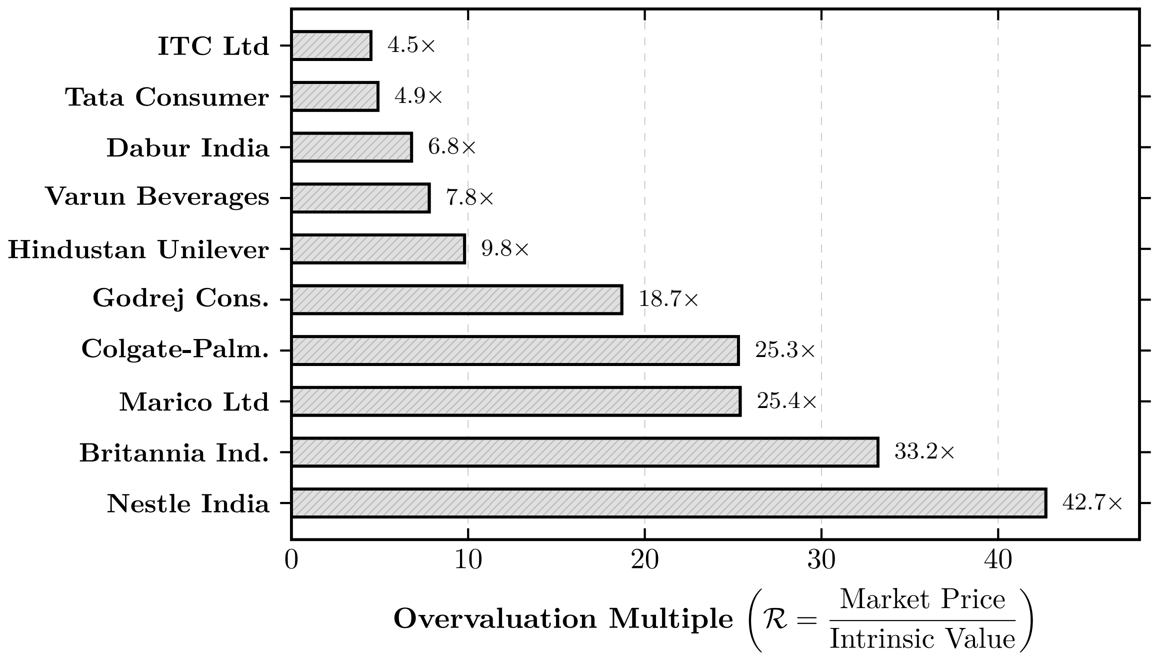

Figure 2: Cross-Sectional Dispersion of Overvaluation Multiples Across Sample FMCG Firms.

3.2 Categorization into Valuation Tiers

To evaluate the relative asset protection available to conservative investors, the universe of ten companies is partitioned into three distinct risk-and-asset backing tiers, shown in Figure 3.

Tier 1: High Asset Backing

Multiple: $4.5\times -6.8\times$

Firms: ITC, Tata Consumer, Dabur India.

Balance Sheet: Substantial tangible asset pool, heavy manufacturing infrastructure, low relative leverage.

Margin of Safety: Highest among FMCG peers.

Tier 2: Intermediate Multiple

Multiple: $7.8\times -18.7\times$

Firms: Varun Beverages, HUL, Godrej Consumer.

Balance Sheet: Hybrid capital structure; massive working capital throughput and operational scale.

Margin of Safety: Moderate asset coverage of market cap.

Tier 3: Extreme Multiples

Multiple: $25.3\times -42.7\times$

Firms: Colgate, Marico, Britannia, Nestle India.

Balance Sheet: Ultra-lean physical asset base, minimal working capital, hyper-efficient ROIC.

Margin of Safety: Negligible liquidation protection.

Figure 3: Classification of FMCG Universe by Asset-Backing Strength and Multiples.

3.3 Margin of Safety Analysis

In classical Graham-and-Dodd value investing, the **Margin of Safety** is defined as the favorable difference between the intrinsic value and the market price paid:

$$\text{Margin of Safety} = \frac{\text{IV} - P_0}{P_0} = \frac{1}{\mathcal{R}} - 1 \tag{11}$$

When applied to the empirical results:

- **ITC Limited** ($\mathcal{R} = 4.5\times$) and **Tata Consumer Products** ($\mathcal{R} = 4.9\times$) present the lowest overvaluation multiples in the entire sector. With Rs. 74,257 Cr and Rs. 20,195 Cr in tangible net assets, these companies offer the strongest balance sheet cushion against downside liquidation shocks.
- **Nestle India Limited** ($\mathcal{R} = 42.7\times$) and **Britannia Industries** ($\mathcal{R} = 33.2\times$) trade at the most extreme multiples relative to physical book equity. An investor acquiring Nestle India at Rs. 1,409 per share is supported by only Rs. 33 in net liquidation equity, implying that over 97.6% of the company's valuation relies entirely on expectations of perpetuity earnings rather than tangible liquidation worth.

Economic Rationale of the Disparity

The large disparity between ITC (4.5×) and Nestle (42.7×) reflects distinct corporate capital architectures. ITC operates capital-intensive agribusiness, paperboards, and hotel divisions requiring extensive physical assets. In contrast, Nestle India functions as an agile branded formulator and marketing powerhouse whose manufacturing footprint is highly optimized, requiring nominal fixed capital per unit of net revenue.

4 Critical Limitations of the NAV Method and Alternative Valuation Frameworks

4.1 Why the Net Asset Value Method Systematically Understates FMCG Value

The uniform finding that **all ten FMCG enterprises trade at significant premiums (**4.5× **to** 42.7×**) above intrinsic liquidation value** is not an artifact of market irrationality, but rather exposes the foundational limitations of static balance sheet valuation when applied to consumer enterprises.

Key Theoretical Insight: The Liquidation Floor

The Net Asset Value (Intrinsic Value) method establishes the **absolute liquidation floor** of a corporate entity—the worst-case cash recovery available upon bankruptcy or immediate dissolution. For a going-concern enterprise compounding returns at high rates of capital efficiency, the balance sheet value represents only the historical physical inputs, not the economic earning power.

The Net Asset Value method is structurally suited for:

1. **Asset-Heavy / Capital-Intensive Sectors:** Real estate investment trusts (REITs), shipping lines, commercial banking, heavy mining, and steel manufacturing, where tangible assets generate direct revenue and liquidation prices are readily observable.

2. **Distressed Liquidation Scenarios:** Situations where an enterprise is experiencing insolvency, operational paralysis, or mandatory debt restructuring.

Conversely, the FMCG sector operates on an **asset-light business model**. The core economic engines that generate enterprise cash flows are uncapitalized off-balance-sheet intangible assets:

- **Brand Equity and Pricing Power:** Established household trademarks (e.g., Maggi, Surf Excel, Aashirvaad, Parachute) command immense consumer loyalty, enabling firms to exercise pricing power and transfer raw material inflationary pressure to retail consumers without sacrificing sales volume.

- **Distribution Infrastructure and Route-to-Market Moats:** Deep logistical networks reaching millions of traditional mom-and-pop retail outlets (kiranas) constitute formidable barriers to entry that cannot be replicated merely by deploying financial capital.

- **Working Capital Architecture:** Leading FMCG corporations frequently operate with **negative net working capital** cycles, effectively funding operational inventory and receivables through interest-free supplier trade credit.

- **High Return on Invested Capital (**ROIC**):** Because physical capital requirements are modest, FMCG firms routinely produce return on equity (ROE) and return on capital employed (ROCE) exceeding 30% to 80%, generating immense economic rent per rupee of book equity.

4.2 Alternative Quantitative Equity Valuation Frameworks

To capture the true intrinsic worth of asset-light, going-concern consumer goods enterprises, quantitative analysts employ models rooted in future earnings, cash flows, and dividends.

4.2.1 Dividend Discount Model (DDM)

Given that established Indian FMCG companies (such as ITC, HUL, Colgate, and Nestle) maintain high dividend payout ratios (often 60% to 90% of net earnings), the **Gordon Growth Model** provides a valuation based on cash distributions to shareholders:

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r_e)^t} = \frac{D_1}{r_e - g} = \frac{D_0(1+g)}{r_e - g} \tag{12}$$

where D_0 is the current annualized dividend per share, D_1 is the anticipated next-period dividend, r_e is the cost of equity derived from the Capital Asset Pricing Model (CAPM: $r_e = R_f + \beta[E(R_m) - R_f]$), and g is the perpetual sustainable dividend growth rate ($g = b \times \text{ROE}$, where b is the retention ratio).

For firms experiencing transient competitive advantages before settling into mature long-term growth, the **Two-Stage Dividend Discount Model** decomposes value into a supernormal growth phase and a stable terminal phase:

$$P_0 = \sum_{t=1}^{T} \frac{D_0(1+g_s)^t}{(1+r_e)^t} + \frac{D_0(1+g_s)^T(1+g_n)}{(r_e - g_n)(1+r_e)^T} \tag{13}$$

where g_s is the initial high growth rate during competitive advantage period T, and g_n is the long-run macroeconomic growth rate.

4.2.2 Price-to-Earnings (P/E) Multiple and Earnings Power Value (EPV)

Under the earnings capitalization approach, share price reflects the multiple that market participants assign to sustainable accounting profits:

$$P_0 = \text{EPS} \times \left(\frac{P}{E}\right)_{\text{target}} \tag{14}$$

From fundamental finance theory, the justified leading P/E multiple is mathematically related to the payout ratio $(1 - b)$ and cost of equity:

$$\left(\frac{P}{E}\right)_{\text{justified}} = \frac{1-b}{r_e - g} \tag{15}$$

Similarly, the **Earnings Power Value (EPV)** framework formalizes the value of current earnings sustainability without speculative growth:

$$\text{EPV} = \frac{\text{Normalized Sustainable Operating Profit} \times (1 - t_{\text{eff}})}{\text{WACC}} \tag{16}$$

where t_{eff} is the effective corporate tax rate and WACC is the Weighted Average Cost of Capital.

4.2.3 Discounted Free Cash Flow to Firm (FCFF)

The **DCF Model** evaluates total operating cash generation independent of capital structure financing decisions:

$$\text{Enterprise Value} = \sum_{t=1}^{T} \frac{\text{FCFF}_t}{(1+\text{WACC})^t} + \frac{\text{FCFF}_{T+1}}{(\text{WACC} - g)(1+\text{WACC})^T} \tag{17}$$

$$\text{Intrinsic Equity Value} = \text{Enterprise Value} - \text{Total Debt} + \text{Cash and Liquid Investments} \tag{18}$$

$$\text{FCFF} = \text{EBIT}(1-t) + \text{D\&A} - \text{CapEx} - \Delta\text{NWC} \tag{19}$$

where D&A denotes Depreciation and Amortization, CapEx represents Capital Expenditures, and ΔNWC is the net investment in Non-Cash Working Capital.

4.3 Methodology Synthesis Table

Table 4 contrasts the Net Asset Value method against earnings- and cash-flow-driven alternatives, highlighting their operational mechanics and sector compatibility.

Valuation Model	Core Governing Formula	Primary Value Driver	Sector Applicability
Net Asset Value	$\text{IV} = \frac{\text{Tangible Assets} - \text{Liabilities}}{\text{Shares}}$	Historical physical book assets	Real estate, shipping, banking, liquidation
Dividend Discount	$P_0 = \frac{D_1}{r_e - g}$	Sustainable cash payouts	Mature, high-dividend FMCG, regulated utilities
Price-to-Earnings	$P_0 = \text{EPS} \times \left(\frac{1-b}{r_e-g}\right)$	Accounting earnings & growth	Broad market benchmarking, consumer goods
Free Cash Flow	$V_0 = \sum \frac{\text{FCFF}_t}{(1+\text{WACC})^t} + \frac{TV}{(1+\text{WACC})^T}$	Operating cash flow generation	Growth FMCG, industrial conglomerates, tech

Table 4: Comparative Overview of Equity Valuation Methodologies.

5 Conclusion and Formula Reference Summary

5.1 Synthesis of Findings and Strategic Investment Implications

The quantitative empirical analysis of ten marquee Indian FMCG corporations illustrates the profound dichotomy between static asset liquidation values and forward-looking market pricing:

1. **Uniform Overvaluation under the Liquidation Metric:** Every company examined trades at a substantial multiple above its net tangible asset worth, ranging from 4.5× (ITC Ltd) to 42.7× (Nestle India Ltd).
2. **Dual Investor Perspectives:**
   - **The Classical Value Investor Perspective:** Under strict Benjamin Graham criteria, none of the ten companies provide an adequate margin of safety based on liquidation assets alone. However, within the peer group, **ITC Ltd** (Rs. 74,257 Cr net assets, 4.5× ratio) and **Tata Consumer Products** (Rs. 20,195 Cr net assets, 4.9× ratio) offer superior downside balance sheet defense.
   - **The Franchise / Quality Investor Perspective:** For growth and franchise investors, the market premium reflects durable competitive advantages (moats), high cash-flow conversion, dependable dividend distributions, and superior Return on Capital Employed (ROCE). Companies like **Nestle India** and **Britannia Industries** trade at $33\times -43\times$ tangible book value precisely because their enterprise value is propelled by high-margin product consumption rather than factory equipment.
3. **Valuation Synthesis:** While the Net Asset Value method fails as an independent estimator of fair market value for asset-light corporations, it performs a vital risk-management function by demarcating the **hard liquidation floor** of the firm in catastrophic downturns.

5.2 Master Reference Table: Valuation Equations

Table 5 provides a consolidated reference of all mathematical formulas implemented throughout this equity valuation project.

Valuation Metric	Mathematical Formulation	Operational Financial Purpose
Average Profit	$\overline{\Pi} = \frac{1}{5}\sum_{t=1}^{5} \Pi_t$	Normalizes historical profitability over a 5-year cycle.
Goodwill	$\text{Goodwill} = \overline{\Pi} \times N_{\text{purchase}} = 3\,\overline{\Pi}$	Quantifies capitalized super-profit earning power.
Tangible Assets	Tangible Assets = Total Assets − Goodwill	Eliminates unamortized/subjective intangibles.
Net Assets (NAV)	Net Assets = Tangible Assets − Total Liab.	Measures residual tangible pool for common equity.
Intrinsic Value (IV)	$\text{IV} = \dfrac{\text{Net Assets Available}}{\text{Number of Equity Shares}}$	Establishes per-share liquidation value floor.
Market Price (P_0)	$P_0 = \dfrac{\text{Market Capitalization}}{\text{Number of Equity Shares}}$	Captures consensus valuation in secondary markets.
Valuation Gap (Δ)	$\Delta = \text{IV} - P_0$	Quantifies absolute divergence from asset backing.
Overvaluation Multiple	$\mathcal{R} = \dfrac{P_0}{\text{IV}} = \dfrac{\text{Market Cap}}{\text{Net Assets}}$	Evaluates relative pricing premium over physical equity.

Table 5: Consolidated Mathematical Formulations for Net Asset Value Equity Valuation.